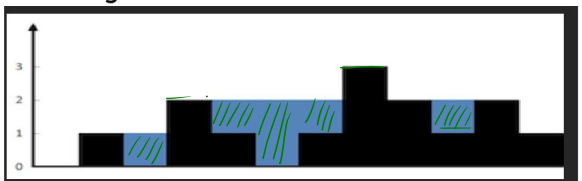
Data Structures and Algorithms
Stack & Queue

Mentor: **Adarsh Gupta**
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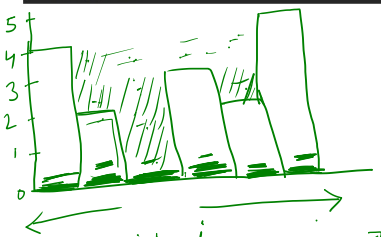
Trapping Rain Water [Hard]

Given n non-negative integers representing an elevation map where the width of each bar is 1, compute how much water it can trap after raining.



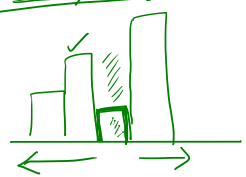
Input: height = [0,1,0,2,1,0,1,3,2,1,2,1]
Output: 6
Explanation: The above elevation map (black section) is represented by array [0,1,0,2,1,0,1,3,2,1,2,1]. In this case, 6 units of rain water (blue section) are being trapped.





✓ [4|2|0|3|2|5] = height

min(leftmax, rightmax) − height[i]



→ O(n)

leftmax[0] = height[0]

✓ leftmax = [4|4|4|4|4|5]

✓ rightmax = [5|5|5|5|5|5]

n = 6 → O(n)

n−1

rightmax[n−1] = height[n−1];

(n−2) → 0

O(3n) ≈ O(n)

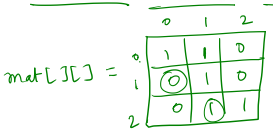
S.C. O(n)

res = 0 + 2 + 4 + 1 + 2 + 0 = 9 ✓

O(n)

The Celebrity Problem [Medium]

A celebrity is a person who is known to all but does not know anyone at a party. A party is being organized by some people. A square matrix mat[][] of size n*n is used to represent people at the party such that if an element of row i and column j is set to 1 it means ith person knows jth person. You need to return the index of the celebrity in the party, if the celebrity does not exist, return -1.



n = 3

0, 1, 2

mat[i][j] = 1 → i knows j person

mat[i][j] = 0 → i does not know j person

Celebrity
- known by everybody
- knows nobody

mat[i][cclib] = 1 ∀ i
mat[cclib][i] = 0 ∀ i

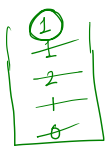
case 1: mat[i][j] = 0 ✓; eliminate j

case 2: mat[i][j] = 1 ✓; eliminate i

n = 3 → 0, 1, 2

mat[0][1] = 1

mat[1][2] = 0



Stack

0, 1, 2

i j
1 0

mat[1][0] = 0

i j

mat[i][j]
mat[2][1] = 1

while (stk.size() > 1) {

3

The Celebrity Problem

Input: mat[][] = [[1, 1, 0],
[0, 1, 0],
[0, 1, 1]]

Output: 1

Explanation: 0th and 2nd person both know 1st person and 1st person does not know anyone. Therefore, 1 is the celebrity person.

Input: mat[][] = [[1, 1],
[1, 1]]

Output: -1

Explanation: Since both the people at the party know each other. Hence none of them is a celebrity person.